

ASP.NET

Unit-I: Introducing .NET and .NET Framework

Comprehensive Study Notes

1. Introducing Microsoft .NET

1.1 What is .NET?

.NET is a **software development platform** created by Microsoft that provides a comprehensive environment for building, deploying, and running applications and web services. It revolutionizes the way developers create applications by offering a unified programming model.

The Move from Workstation to Distributed Computing

Era	Characteristics	Limitations
Workstation Era	Standalone applications, Local data storage	No sharing, Limited resources
Client-Server Era	Centralized database, Client applications	Tight coupling, Scalability issues
Distributed Era	Multiple servers, Load balancing	Complex management, Platform dependency
Internet Era	Global access, Web services	Security concerns, Connectivity issues

The Internet Factor

The internet has transformed application development in several ways:

1. **Global Reach:** Applications can be accessed from anywhere
2. **Platform Independence:** Works across different operating systems
3. **Real-time Communication:** Instant data exchange
4. **Service-oriented Architecture:** Components can be consumed as services

2. Importance of .NET Platform

2.1 OS Neutral Environment

- .NET applications are compiled to **Intermediate Language (IL)**
- IL is platform-independent
- **Common Language Runtime (CLR)** executes IL on any supported OS
- .NET Core supports Windows, Linux, and macOS

2.2 Device Independence

Same code can run on different devices including desktops, laptops, tablets, and mobile phones.

2.3 Wide Language Support

.NET supports multiple programming languages including VB.NET, C#, F#, and Managed C++, allowing developers to use their language of choice.

2.4 Internet-based Component Services

- **Web Services:** SOAP-based services
- **REST APIs:** HTTP-based services
- **Microservices:** Independent deployable services
- **Component Reusability:** Build once, use everywhere

3. .NET Framework Architecture

3.1 Common Language Runtime (CLR)

The CLR is the **execution engine** of the .NET Framework. It manages the execution of .NET programs and provides core services.

Functions of CLR:

1. **Code Management and Execution** - Loads and executes managed code
2. **Security Support** - Enforces code access security (CAS)
3. **Error Handling** - Structured exception handling via try-catch-finally
4. **Garbage Collection** - Automatic memory management

Error Handling Example:

```
Try
    ' Risky code
Catch ex As DivideByZeroException
    ' Handle specific exception
Catch ex As Exception
    ' Handle any exception
Finally
    ' Cleanup code
End Try
```

3.2 .NET Framework Class Libraries (FCL)

The FCL provides a **huge collection of reusable classes** organized into namespaces.

Category	Key Classes	Purpose
System Classes	Object, String, Int32, DateTime, Math, Console, Environment	Fundamental data types, basic operations
Data Classes	SqlConnection, DataSet, DataTable, SqlCommand	Database connectivity (ADO.NET)

Windows Forms	Form, Button, TextBox, Label, DataGridView	Desktop GUI applications
Drawing Classes	Graphics, Pen, Brush, Bitmap, Color	2D graphics and image manipulation
Web Classes	Page, Control, HttpRequest, HttpResponse, SessionState	Web application development

3.3 Key Components of .NET Framework

MSIL / IL (Intermediate Language)

.NET source code is compiled into MSIL (Microsoft Intermediate Language), which is CPU-independent. IL is later converted to native code by the JIT compiler.

JIT (Just-In-Time) Compiler

Converts IL into native machine code at runtime. Types: Pre-JIT, Econo-JIT, and Normal-JIT.

BCL / FCL (Base Class Library / Framework Class Library)

A comprehensive collection of reusable types including classes, interfaces, and value types that form the foundation of .NET applications.

CTS (Common Type System)

Defines how types are declared, used, and managed in the runtime. Ensures type consistency across all .NET languages.

CLS (Common Language Specification)

A set of rules and guidelines that ensures language interoperability. Any language that follows CLS can use components from any other CLS-compliant language.

GC (Garbage Collector)

Automatic memory management system that tracks and reclaims memory occupied by objects no longer in use.

4. Languages in .NET

- **VB.NET** - Evolution of Visual Basic, fully object-oriented
- **C#** - Modern, type-safe language with C-like syntax
- **F#** - Functional-first programming language
- **Managed C++** - C++ with .NET extensions for CLR

5. ASP.NET

5.1 Introduction

ASP.NET is a web application framework developed by Microsoft for building dynamic web pages, web applications, and web services. It is part of the .NET platform.

ASP.NET vs Classic ASP

Feature	Classic ASP	ASP.NET
Programming Model	Interpreted scripting	Compiled code
Language Support	VBScript, JScript	VB.NET, C#, F#, etc.
Code Separation	Mixed HTML and script	Code-behind separation
State Management	Limited	Comprehensive (ViewState, Session, etc.)
Error Handling	Basic	Structured exception handling
Performance	Interpreted (slower)	Compiled (faster)

5.2 Key Features of ASP.NET

- **Compiled Execution** - Faster performance through JIT compilation
- **Multi-language Support** - Can use any .NET language
- **Server Controls** - Rich set of reusable UI components
- **State Management** - ViewState, Session, Application, Cache
- **Security** - Authentication, authorization, and code access security
- **ADO.NET** - Data access framework for databases
- **Caching** - Output, fragment, and data caching

- **Mobile Support** - Mobile controls for mobile device rendering

5.3 ASP.NET Server Controls

Control Type	Examples	Key Features
Web Server Controls	Label, TextBox, Button, GridView, Calendar	Rich functionality, AutoPostBack, ViewState
HTML Server Controls	HtmlInput, HtmlTable, HtmlAnchor	HTML compatibility, Simple rendering
Validation Controls	RequiredFieldValidator, RangeValidator, CompareValidator	Client/Server validation, Display errors
Data Controls	GridView, DataList, Repeater	Data binding, Templates, Paging

5.4 ASP.NET Infrastructure

State Management

ViewState (page-level), Session (user-level), Application (application-level), Cache (performance).

Security Infrastructure

Authentication (Windows, Forms, Passport), Authorization, and Impersonation.

Caching

Output Caching (entire page), Fragment Caching (user controls), and Data Caching (programmatic).

Configuration

XML-based configuration files (web.config) for application settings.

5.5 Web Form Example


```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb"
Inherits="_Default" %>
<html>
<head><title>My Web Form</title></head>
<body>
    <form id="form1" runat="server">
        <asp:Label ID="lblMessage" runat="server" Text="Hello ASP.NET">
</asp:Label>
    </form>
</body>
</html>
```

6. .NET Core vs .NET Framework

Feature	.NET Framework	.NET Core
Platform	Windows only	Cross-platform (Windows, Linux, macOS)
Open Source	Partial	Fully open source
Deployment	Machine-wide install	Side-by-side, self-contained
Performance	Good	High performance, optimized
Application Types	Desktop, Web, Enterprise	Web, Cloud, Microservices, Console
API Surface	Full Windows API	Modern, modular APIs

7. Quick Reference Tables

.NET Framework Versions

Version	Year	Key Features
1.0	2002	Initial release, CLR 1.0
1.1	2003	ASP.NET mobile controls, ADO.NET improvements
2.0	2005	Generics, Master Pages, Membership providers
3.0	2006	WPF, WCF, WF, CardSpace
3.5	2007	LINQ, AJAX, Entity Framework
4.0	2010	Parallel Library, Dynamic Language Runtime
4.5	2012	Async/Await, Web Sockets
4.8	2019	Latest .NET Framework version

Common Language Runtime Components

Component	Function
Class Loader	Loads classes at runtime
JIT Compiler	Converts IL to native code
Garbage Collector	Automatic memory management
Security Engine	Enforces code access security
Exception Handler	Manages structured exceptions
Thread Support	Manages threading and synchronization

Key Namespaces for Web Development

```
Imports System.Web           ' Web application basics
Imports System.Web.UI       ' Web controls
Imports System.Web.UI.WebControls ' Server controls
Imports System.Web.UI.HtmlControls ' HTML controls
Imports System.Data         ' Database access
Imports System.Data.SqlClient ' SQL Server access
Imports System.IO           ' File operations
Imports System.Net          ' Network operations
```

Important Questions for Unit I

Short Answer Questions (2 marks)

1. What is .NET Framework?
2. Define CLR and its role.
3. What is Garbage Collection?
4. What are managed and unmanaged code?
5. List any four components of .NET Framework.
6. What is JIT Compiler?

Medium Answer Questions (5 marks)

1. Explain the architecture of .NET Framework with a diagram.
2. What are the functions of CLR? Explain in detail.
3. Describe the importance of .NET platform.
4. What is .NET Framework Class Library? Explain its components.

Long Answer Questions (12 marks)

1. Draw and explain the .NET Framework architecture with all its components.
2. Explain the Common Language Runtime and its services in detail.
3. Compare workstation, client-server, and distributed computing models in the context of .NET.
4. What are the features of .NET Framework? Discuss ASP.NET web forms and services.

"The .NET Framework provides a complete, consistent programming model for building applications that have visually stunning user experiences and seamless and secure communication." - Microsoft